Chapter 4 Basic equations of steady total flow

Section 1 Method of total flow analysis

Section 2 Continuity equation

☞ Section 3 Bernoulli equation for steady total flow

Section 4 Momentum equation for steady total flow

Chapter summary

Section 3 Bernoulli equation for steady total flow

☞ 1. Bernoulli equation for element flow of real fluid

2. Bernoulli equation for steady total flow

3. Water head line

4. Extension of Bernoulli equation

Section 3 Bernoulli equation for steady total flow

1. Bernoulli equation for element flow of real fluid

Bernoulli equation of ideal fluid: $z+\dfrac{p}{\rho g}+\dfrac{u^2}{2g}=C$

Considering head loss, let:

$h_w{}'$ ——The mechanical energy loss per unit weight of viscous fluid flowing from the first flow section to the second flow section, called the head loss of element flow.

From the principle of energy conservation, the Bernoulli Equation for element flow of viscous fluid, reads:

$$z_1+\frac{p_1}{\rho g}+\frac{u_1^2}{2g}=z_2+\frac{p_2}{\rho g}+\frac{u_2^2}{2g}+h_w{}'$$

Excise 1: The principle of Pitot-tube velocity measurement

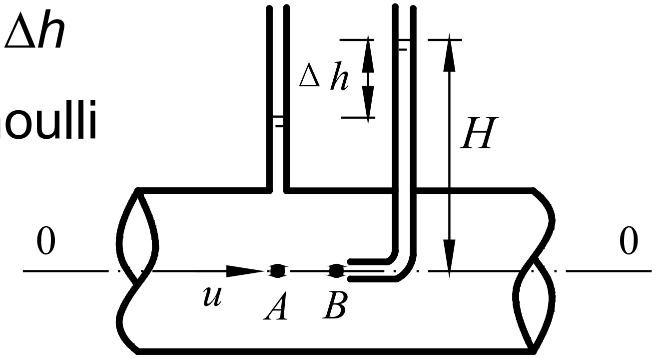


Fig 4-6. Pitot tube speed measurement principle

Question: Calculate the u at point A according to Δh

Solution: Select the datum; according to the Bernoulli equation for viscous fluid element flow, we have:

$$z_{\mathrm{A}}+\frac{p_{\mathrm{A}}}{\rho g}+\frac{u_{\mathrm{A}}^{2}}{2g}=z_{\mathrm{B}}+\frac{p_{\mathrm{B}}}{\rho g}+0+h_{\mathrm{w}}'$$

Set $u_A = u$ and $h_{\mathrm{w}}' = \zeta\dfrac{u^2}{2g}$

where ζ is called head loss factor and can be determined by experiment. The value of ζ is greater than 0 and close to 0.

Therefore: $$u=\frac{1}{\sqrt{1-\zeta}}\sqrt{2g\left[\left(z_{\mathrm{B}}+\frac{p_{\mathrm{B}}}{\rho g}\right)-\left(z_{\mathrm{A}}+\frac{p_{\mathrm{A}}}{\rho g}\right)\right]}=\frac{1}{\sqrt{1-\zeta}}\sqrt{2g\Delta h}$$

Letting $\dfrac{1}{\sqrt{1-\zeta}}=c$ $\longrightarrow$ $u=c\sqrt{2g\Delta h}$

- c is called the Pitot tube factor.
- Related to head loss in energy conversion and the structure of Pitot tube
- Determined by experiments, the value is close to 1.0

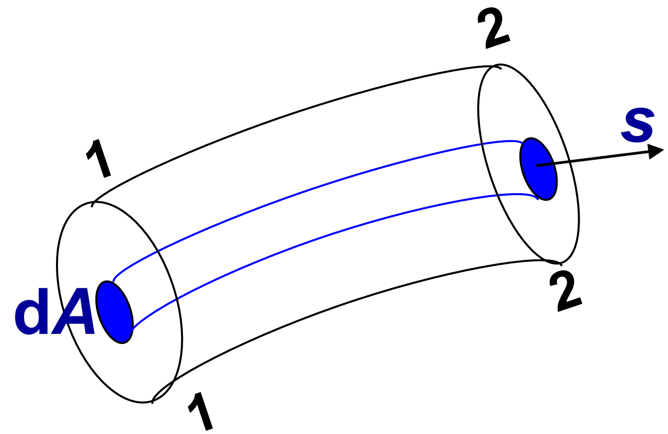
Section 3 Bernoulli equation for steady total flow

1. Bernoulli equation for element flow of real fluid
2. Bernoulli equation for steady total flow
3. Water head line
4. Extension of Bernoulli equation

2. Bernoulli equation for steady total flow

(1) Derivation of the equation

$$z_1+\frac{p_1}{\rho g}+\frac{u_1^2}{2g}=z_2+\frac{p_2}{\rho g}+\frac{u_2^2}{2g}+h_{\mathrm{w}}'$$



- Let the flow rate be $\mathrm{d}q_V=u_1\mathrm{d}A_1=u_2\mathrm{d}A_2$.
- Multiply $\rho g\mathrm{d}q_V$ at both sides of the above equation for element flow
- The energies of fluids flowing through the two cross sections of the element flow, per unit time, satisfy:

$$(z_1+\frac{p_1}{\rho g}+\frac{u_1^2}{2g})\rho g u_1\mathrm{d}A_1=(z_2+\frac{p_2}{\rho g}+\frac{u_2^2}{2g})\rho g u_2\mathrm{d}A_2+h_{\mathrm{w}}'\rho g\mathrm{d}q_V$$

Integrating the energy equation on the cross section of total flow:

$$\int_{A_1}\left(z_1+\frac{p_1}{\rho g}+\frac{u_1^2}{2g}\right)\rho g u_1 \mathrm{d}A_1=\int_{A_2}\left(z_2+\frac{p_2}{\rho g}+\frac{u_2^2}{2g}\right)\rho g u_2 \mathrm{d}A_2+\int_Q h_{\mathrm{w}}{}'\rho g \mathrm{d}q_V$$

1) Potential energy: On a section of uniform flow or gradually-varied flow, we have

$$z+\frac{p}{\rho g}=C$$

Then $\rho g\int_A\left(z+\frac{p}{\rho g}\right)u\mathrm{d}A=\rho g\left(z+\frac{p}{\rho g}\right)\int_A u\mathrm{d}A=\left(z+\frac{p}{\rho g}\right)\cdot\rho g q_V$

2) Kinetic energy: $\rho g\int_A\frac{u^3}{2g}\mathrm{d}A=\frac{\rho g}{2g}\alpha v^3 A=\frac{\alpha v^2}{2g}\rho g q_V$

3) Energy loss: $\int_{q_V} h_{\mathrm{w}}{}'\rho g \mathrm{d}q_V=h_{\mathrm{w}}\rho g q_V$

Bernoulli 's equation for steady total flow of real fluids (of unit weight)

$$z_1+\frac{p_1}{\rho g}+\frac{\alpha_1 v_1^2}{2g}=z_2+\frac{p_2}{\rho g}+\frac{\alpha_2 v_2^2}{2g}+h_{\mathrm{w}}$$

$$z_1 + \frac{p_1}{\rho g} + \frac{\alpha_1 v_1^2}{2g} = z_2 + \frac{p_2}{\rho g} + \frac{\alpha_2 v_2^2}{2g} + h_{\mathrm{w}}$$

(2) Application conditions of Bernoulli Equation for steady total flow

1) Steady flow；

2) Incompressible fluid；

3) Gravity being the sole mass force；

4) Two flow cross sections selected must be gradually-varied or uniform flow sections, while flows between the two sections can be rapidly-varied ones;

5) Rate of total flow remains unchanged along the way;

6) There is no energy input or output in the total flow except for the head loss between the two flow sections;

7) Each term is the average energy of fluid per unit weight (i.e. specific energy). The Bernoulli equation for the total weight of fluid should be multiplied by $\rho g q_V$.

(3) Application of the Bernoulli equation for steady total flow

"Three selections & one list"

1) Select datum plane

- E.g., the centroid of the flow section ($z = 0$), the free surface ($p = 0$)

2) Select calculation section

- Uniform or gradually-varied flow sections; sections with more quantities

3) Select calculation point

- Pipe flow usually selected on the section center
- Open channel flow usually selected on the free surface.

4) List the Bernoulli equation and solve：

- For the same equation, the same pressure standard must be adopted
- The continuity equation can be incorporated

(4) Physical and geometric meaning of Bernoulli Equation for steady total flow

$$z_1 + \frac{p_1}{\rho g} + \frac{\alpha_1 v_1^2}{2g} = z_2 + \frac{p_2}{\rho g} + \frac{\alpha_2 v_2^2}{2g} + h_{\mathrm{w}}$$

The physical and geometric meanings are the same as those for element flow Bernoulli equation, but in an average sense.

	Physical meaning	Geometric meaning
z —	Specific potential energy at the calculation point on the total-flow cross section	Elevation head
$\frac{p}{\rho g}$ —	Specific pressure energy	Pressure head
$\frac{u^2}{2g}$ —	Specific kinetic energy	Flow head
$z+\frac{p}{\rho g}$ —	Total specific potential energy	Hydraulic (piezometric) head
$z+\frac{p}{\rho g}+\frac{u^2}{2g}$ —	Total specific energy	Total head

(5) Differences between the Bernoulli equations for total flow and element flow?

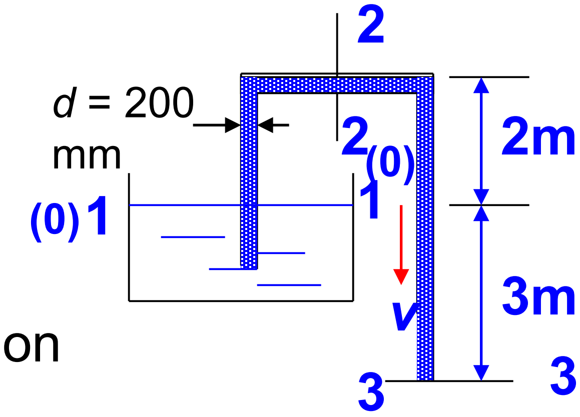
1) Replacing the point velocity u for element flow with the average velocity of a section in total flow, i.e. v

2) Replacing the head loss of element flow, i.e. $h'_{w1.2}$, with the average head loss h_w

Example 1 : As shown in the figure of the siphon discharge, given the losses on section 1,2 and 2,3 are $h_{w1,2}=0.6v^2/(2g)$ and $h_{w2,3}=0.5v^2/(2g)$, calculate the average pressure on section 2.

Solution: Set 0-0 as the datum and let $\alpha_1 = \alpha_2 = 1$.

The Bernoulli equation for sections 1 & 2 reads:



$$0+0+0=2+\frac{p_2}{\rho g}+\frac{v_2^2}{2g}+0.6\frac{v_2^2}{2g} \qquad \text{(a)}$$

With $v_2=v_3=v$ (because $d_2=d_1=d$), the Bernoulli equation for sections 1 & 3 reads:

$$0+0+0=-3+0+\frac{v_3^2}{2g}+0.6\frac{v_3^2}{2g}+0.5\frac{v_3^2}{2g} \qquad \text{(b)}$$

So: $$\frac{v_3^2}{2g}=\frac{v_2^2}{2g}=\frac{v^2}{2g}=1.43\ \mathrm{m}$$

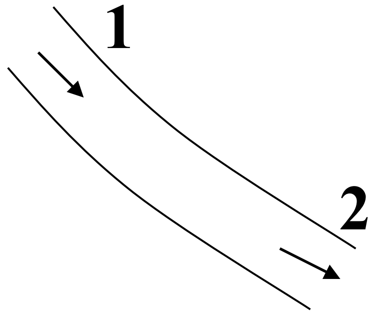
Substituting gives: $\frac{p_2}{\rho g}=-4.29\ \mathrm{m},\ p_2=-9.8\times 4.29=-42.04\ \mathrm{kPa}$

- Solution shows that the relative pressure at the top of the siphon is negative, i.e. vacuum happening.
- To mitigate cavitation, the height of siphon top (i.e. suction high) needs to be controlled to prevent the formation of excessive vacuum.

1. Take two thin paper sheets and hold them parallelly in the hand. Blow air using your mouth into the gap of the paper sheet. Do the paper sheets keep stationary, get close, or separate?

Answer: Paper sheets get close, because flow rate increases and pressure drops

2. Water flows in equal-diameter inclined pipe, with higher point 1 and lower point 2 indicated. Compare the flow velocities at points 1 and 2, when pressures satisfy: $p_1>p_2$; $p_1=p_2$



$$z_1+\frac{p_1}{\rho g}+\frac{\alpha_1 v_1^2}{2g}=z_2+\frac{p_2}{\rho g}+\frac{\alpha_2 v_2^2}{2g}+h_w$$

- $p_A>p_B$: $v_2 > v_1$
- $p_A=p_B$: $v_2 > v_1$

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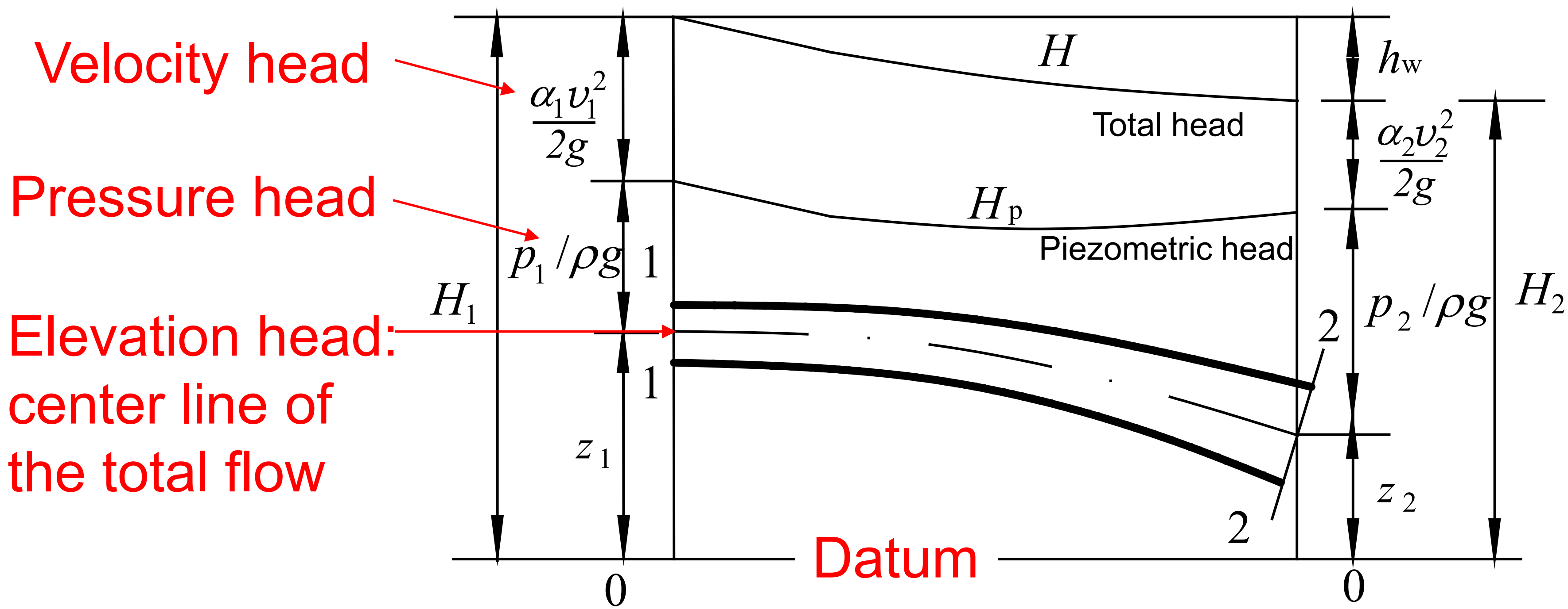
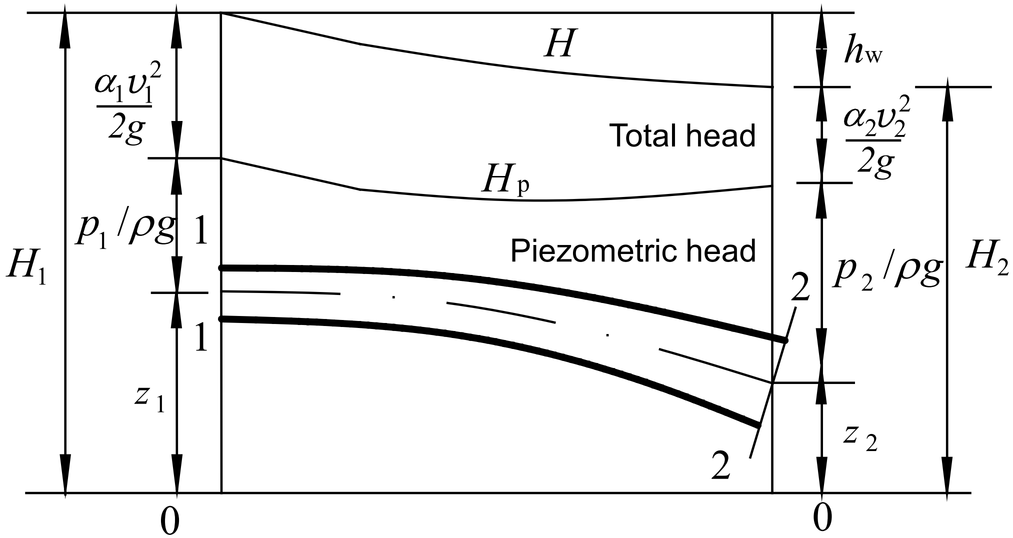
3. Water head line

Since all the terms in the Bernoulli equation have the unit of length, we can use geometric segments to describe the energy variation along flow direction.



Piezometric head：测压管水头

3. Water head line



Hydraulic slope J: The mean head loss per unit length of flow,

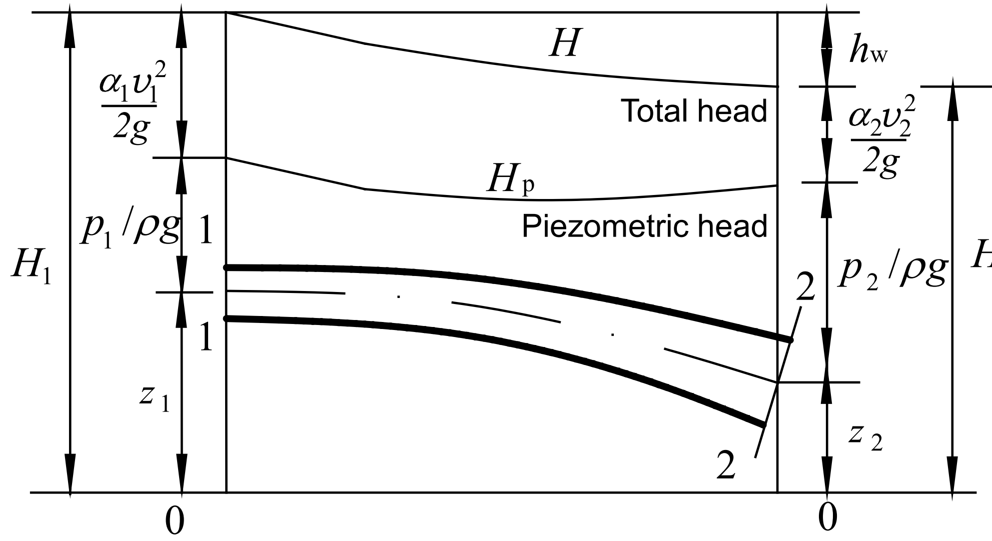
水力坡降

$$J = \frac{\mathrm{d}h_w}{\mathrm{d}s} = -\frac{\mathrm{d}H}{\mathrm{d}s} > 0$$

Slope of piezometric head J_P: Drop of piezometric head per unit length of flow (measured by piezometric tube),

$$J_p = \frac{-\mathrm{d}(z + \dfrac{p}{\rho g})}{\mathrm{d}s}$$

Some notes:



1) The total head line of ideal fluid flow is horizontal;

2) Total head line of actual fluid flow is a descending curve;

3) The piezometric head line can be ascending, descending or horizontal;

4) For uniform flow, the total head line is parallel to the piezometric head line;

5) The difference between the total head line and the piezometric head line corresponds to the velocity head of a certain section.

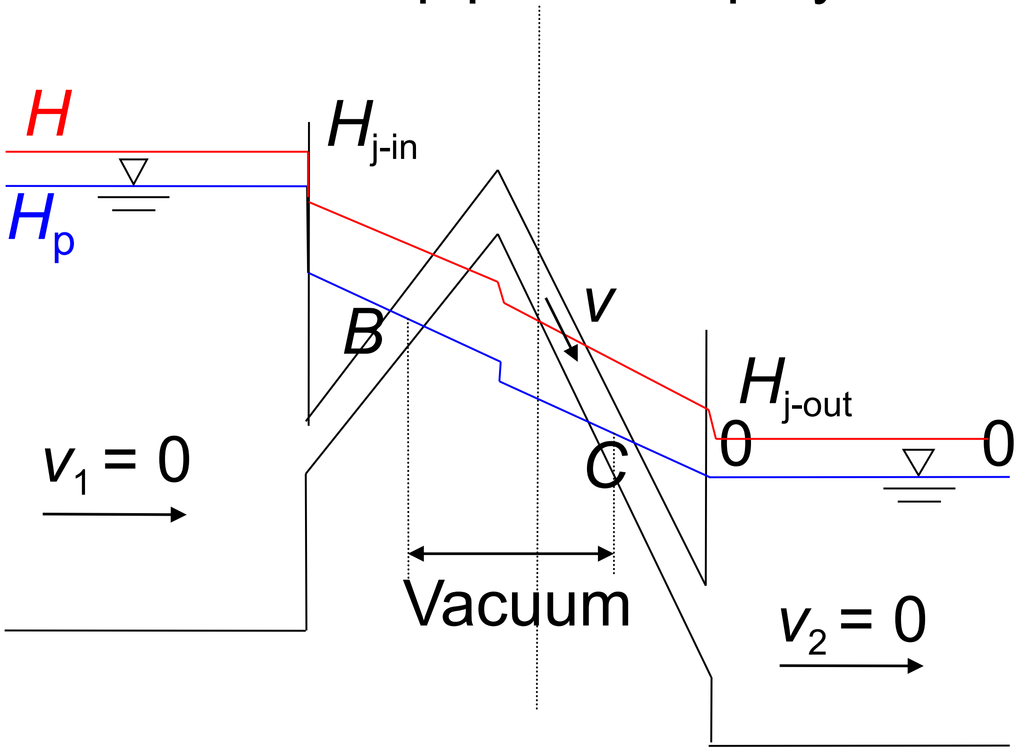
Application of water head line

- Prediction of pipeline vacuum range
- Determination of the highest location of pipeline deployment

In segment BC,
comparing *Hp* and
elevation head gives

$$z+\frac{p}{\rho g}\le z$$

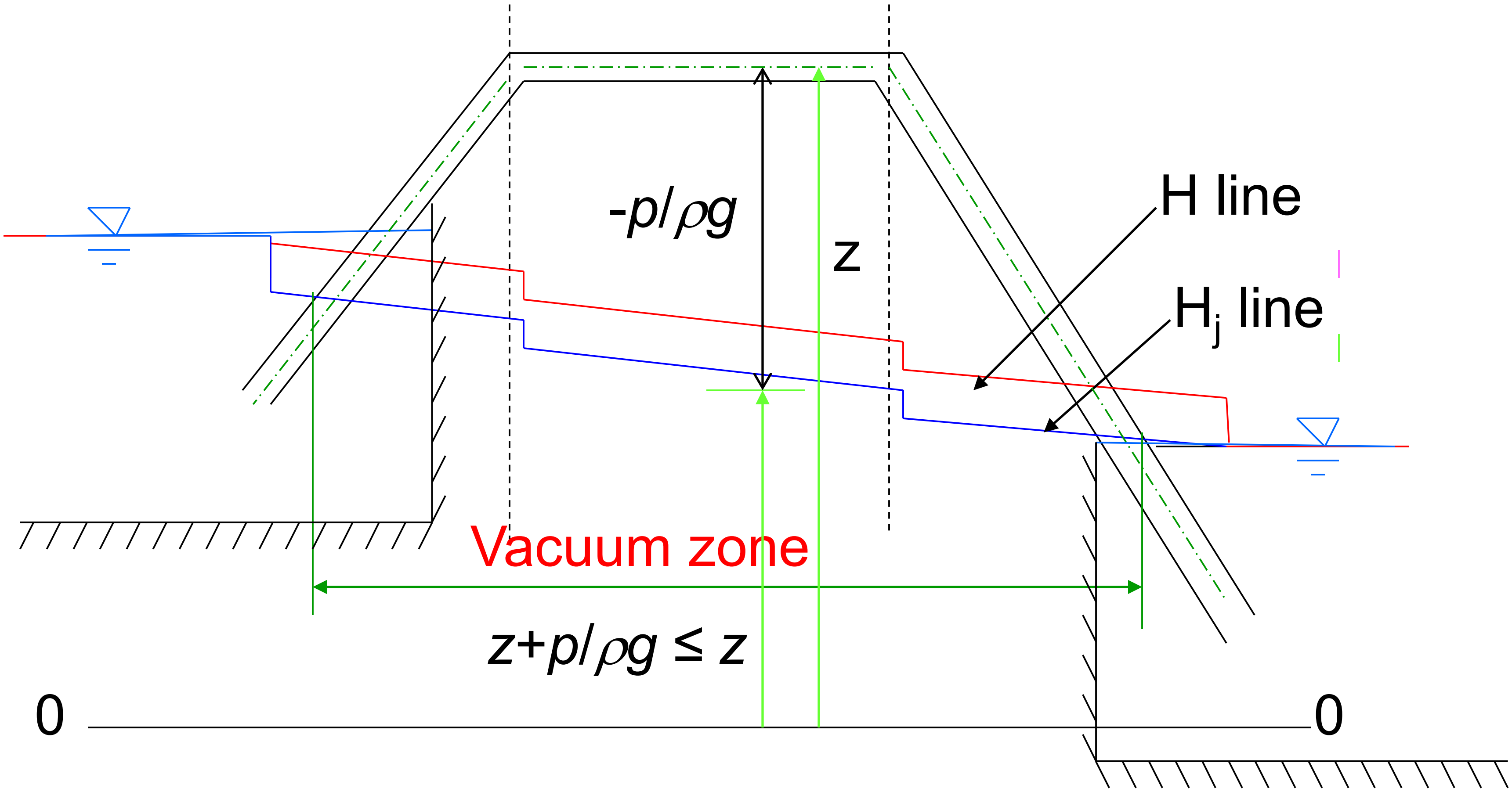
$$\therefore \frac{p}{\rho g}<0$$



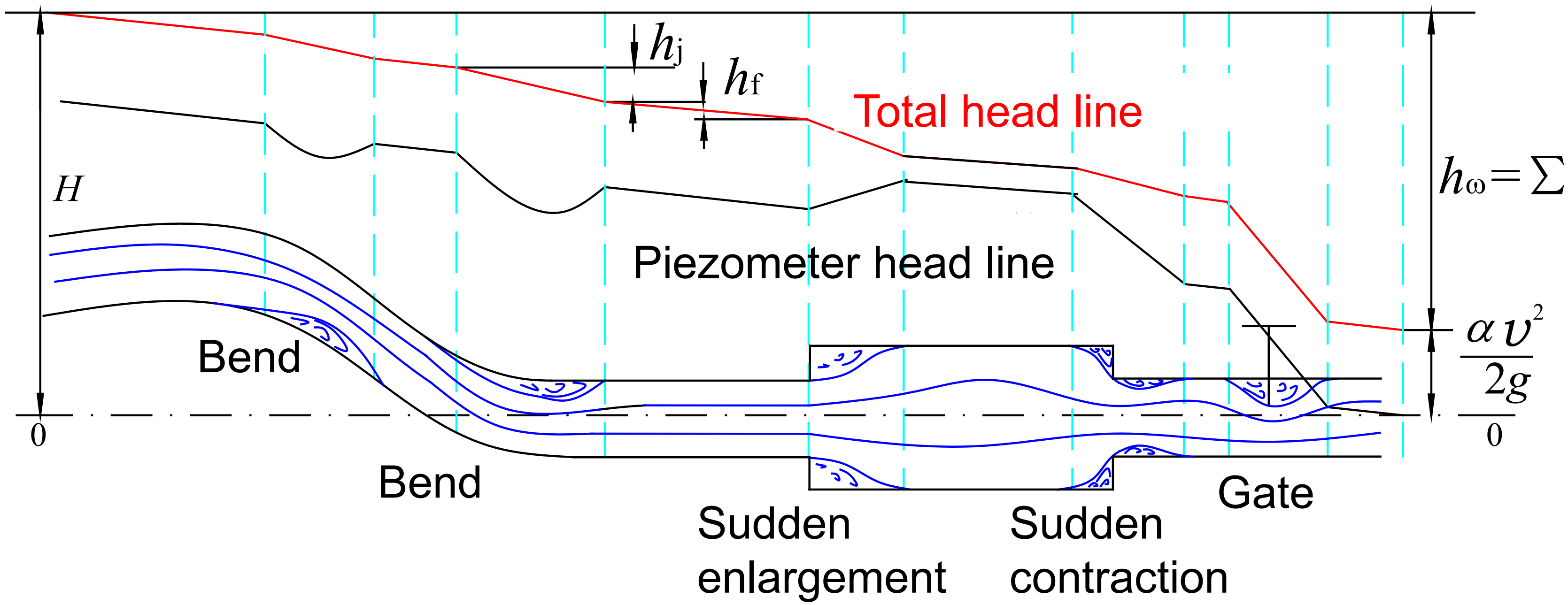
This implies that vacuum occurs in BC.

The head line and

the vacuum zone of siphon pipe (where $z+p/\rho g \le z$)



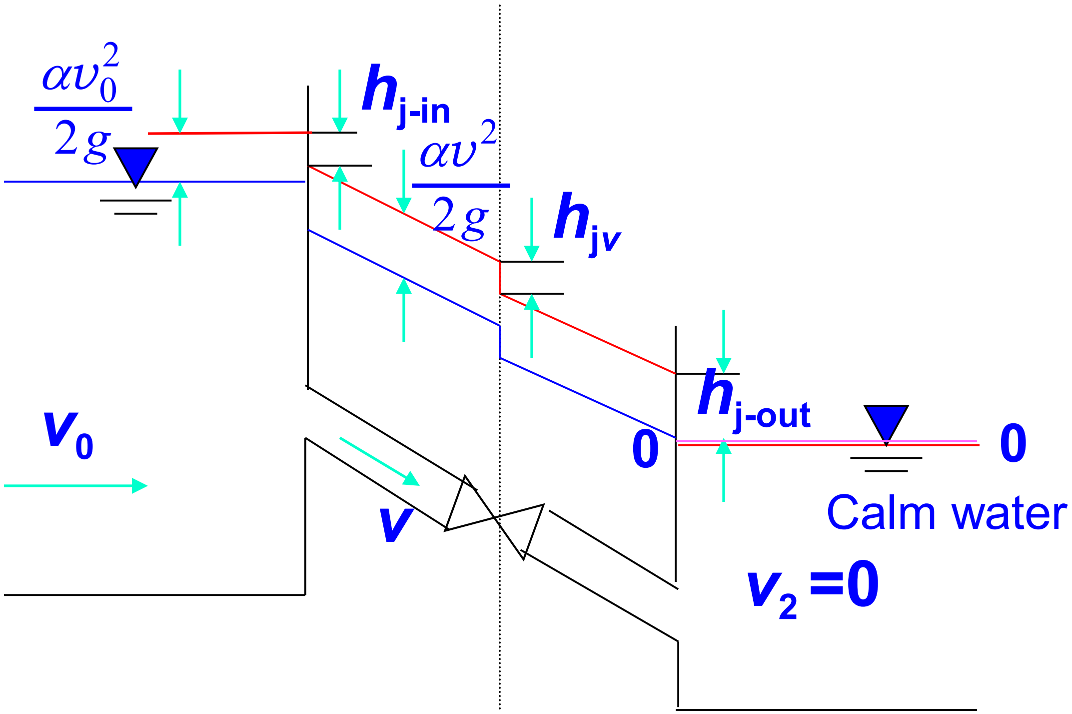
Headlines under different solid boundaries



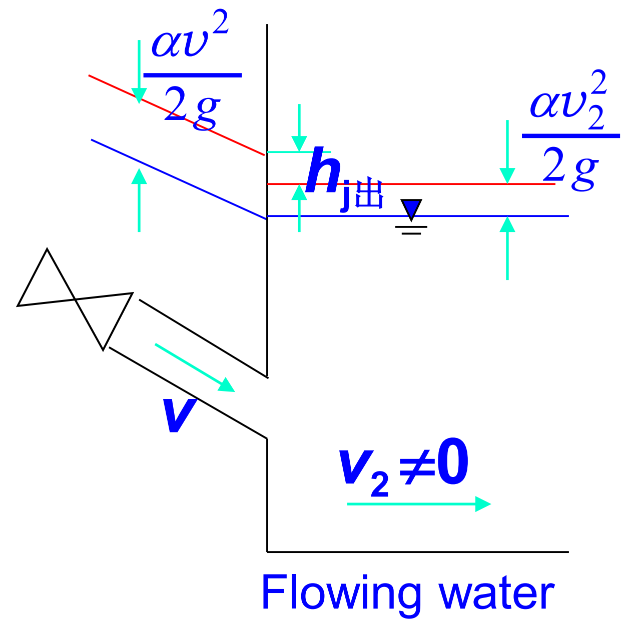
- When the outlet is free outflow, the end of the Hp (or denoted by H_j) line locates on the pipe axis of the outlet section

Headlines under different solid boundaries

- Zero flow rate in the downstream reservoir



- Non-zero flow rate in the downstream reservoir



- The piezometer and total head lines at the outlet coincide with the water level of the downstream reservoir
- The total head line at the outlet of the pipeline is higher than the downstream water level

Questions

1. For a horizontal pressure pipe flow without head loss, what are the conditions for the drop, rise and keeping horizontal of the piezometer head line?

2. What is the difference between the total head line and the piezometer (or hydraulic) head line ?

Questions

1. For a horizontal pressure pipe flow without head loss, what are the conditions for the drop, rise and keeping horizontal of the piezometer head line?

Drop: Pressure decreases along the path of flow motion, e.g. contraction pipe

Keeping horizontal: Pressure remains unchanged along flow motion, e.g. pipe diameter not changing

Rise: Pressure increases along flow motion, e.g. pipe with enlarged section

2. What is the difference between the total head line and the piezometer (or hydraulic) head line?

- The total head line is a curve representing the variation of the total head, i.e. $z+\frac{p}{\rho g}+\frac{\alpha v^2}{2g}$, along the path of flow motion
- The piezometer head line is a curve representing the variation of the piezometric head, i.e. $z+\frac{p}{\rho g}$, along the path of flow motion

Section 3 Bernoulli equation for steady total flow

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4. Extension of Bernoulli equation

(1) Bernoulli equation accounting for flow division or confluence along the path of flow motion

1) Flow division

The Bernoulli equation applies between Sections 1 & 2 or Sections 1 & 3, which gives:

$$z_1+\frac{p_1}{\rho g}+\frac{\alpha_1 v_1^2}{2g}=z_2+\frac{p_2}{\rho g}+\frac{\alpha_2 v_2^2}{2g}+h_{w1,2}$$

Or $$z_1+\frac{p_1}{\rho g}+\frac{\alpha_1 v_1^2}{2g}=z_3+\frac{p_3}{\rho g}+\frac{\alpha_3 v_3^2}{2g}+h_{w1,3}$$

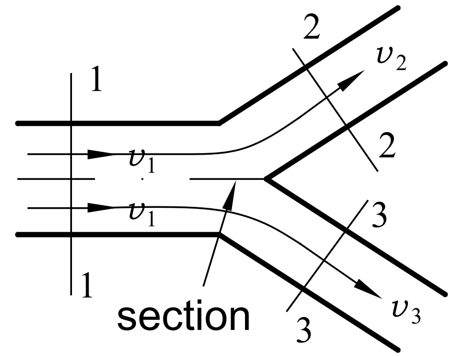


图4-11 Division

2) Flow confluence

The conservation of total energy for total flow gives:

$$\rho g q_{V1}(z_1+\frac{p_1}{\rho g}+\frac{\alpha_1 v_1^2}{2g})+\rho g q_{V2}(z_2+\frac{p_2}{\rho g}+\frac{\alpha_2 v_2^2}{2g})$$

$$=\rho g q_{V3}(z_3+\frac{p_3}{\rho g}+\frac{\alpha_3 v_3^2}{2g})+\rho g q_{V1} h_{w1,3}+\rho g q_{V2} h_{w2,3}$$

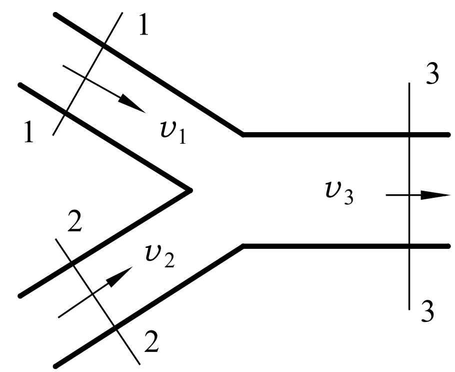


图4-12 Confluence

(2) Bernoulli equation accounting for energy input or output along the path of flow motion

In a flow, if there is output (e.g. hydraulic turbine) or input (e.g. pump or fan) of mechanical energy, the Bernoulli equation can be modified as follows:

$$z_1 + \frac{p_1}{\rho g} + \frac{\alpha_1 v_1^2}{2g} \pm H = z_2 + \frac{p_2}{\rho g} + \frac{\alpha_2 v_2^2}{2g} + h_{\mathrm{w}1,2}$$

where,

+*H*: The energy obtained by fluid of unit weight from fluid machinery like water pump, fan, etc.

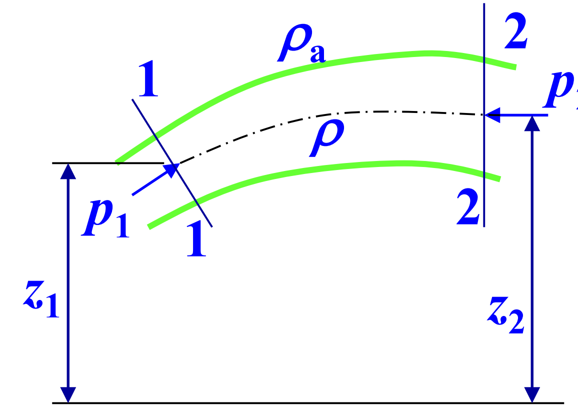
-*H*: The energy lost by fluid of unit weight passing through fluid machinery like turbine.

(3) Bernoulli equation for airflow

A gas can be regarded as <u>incompressible</u> when its flowing velocity is less than about 50 m/s.

Letting $\alpha_1 = \alpha_2 = 1$, the Bernoulli equations at Sections 1-1 and 2-2 reads,

$$z_1 + \frac{p_{1abs}}{\rho g} + \frac{v_1^2}{2g} = z_2 + \frac{p_{2abs}}{\rho g} + \frac{v_2^2}{2g} + h_w$$



In the calculation of airflow, the above equation is usually expressed in the form of pressure, as

$$\rho g z_1 + p_{1abs} + \frac{\rho v_1^2}{2} = \rho g z_2 + p_{2abs} + \frac{\rho v_2^2}{2} + p_w \quad (1)$$

where p_w is the pressure loss with $p_w = \rho g h_w$.

Since $P_{1abs} = p_1 + p_a$, we have $P_{2abs} = p_2 + p_a - \rho_a g (z_2 - z_1)$

where p_a is the atmospheric pressure at z_1, and $p_a - \rho_a g(z_2 - z_1)$ is the atmospheric pressure at z_2 elevation. Substitution gives:

$$p_1 + \frac{\rho v_1^2}{2} + (\rho_a - \rho) g (z_2 - z_1) = p_2 + \frac{\rho v_2^2}{2} + p_w \quad (2)$$

——Bernoulli equation for airflow

Physical meaning of each term in the Bernoulli equation for incompressible gas flow

$$p_1 + \frac{\rho v_1^2}{2} + (\rho_a - \rho) g (z_2 - z_1) = p_2 + \frac{\rho v_2^2}{2} + p_w \qquad (2)$$

p_1. p_2: static pressure

$\frac{\rho v_1^2}{2}, \frac{\rho v_2^2}{2}$: dynamic pressure

$(\rho_a - \rho)g$: effective buoyancy force per unit volume of gas

$(z_2 - z_1)$: distance of gas rising along the direction of buoyancy

$(\rho_a - \rho)g(z_2 - z_1)$: potential energy of gas per unit volume on Section 1-1 relative to Section 2–2, called potential pressure.

$$p_1+\frac{\rho v_1^2}{2}+(\rho_a-\rho)g(z_2-z_1)=p_2+\frac{\rho v_2^2}{2}+p_w$$

- When the density of the airflow equal to that of ambient air (i.e. $\rho_a = \rho$), or when the heights of two calculation points are the same (i.e. $z_2 = z_1$), the potential pressure $(\rho_a - \rho)g(z_2 - z_1)$ is zero. Eq. (2) can be simplified to:

$$p_1+\frac{\rho v_1^2}{2}=p_2+\frac{\rho v_2^2}{2}+p_w$$

where the sum of static and dynamic pressures is the total pressure.

- When the density of the airflow is much larger than that of the ambient air (i.e. $\rho >> \rho_a$), ρ_a in Eq. (2) is negligible and hence the local atmospheric pressure at each point is the same. This is equivalent to the total liquid flow. Eq. (2) can be simplified to

$$p_1+\frac{\rho v_1^2}{2}-\rho g(z_2-z_1)=p_2+\frac{\rho v_2^2}{2}+p_w$$

Divided by ρg, gives: $z_1+\frac{p_1}{\rho g}+\frac{v_1^2}{2g}=z_2+\frac{p_2}{\rho g}+\frac{v_2^2}{2g}+h_w$

Thus, for the total flow of liquid, no matter pressures p_1 and p_2 are absolute pressure or gauge pressure, the form of Bernoulli equation keeps the same.

Example 1：Jet device（射流器）

Q: Given flow rates q_{V1} = 0.004 m³/s, q_{V2} = 0.0005 m³/s, h = 5m, D=0.05m. Ignoring head loss, calculate the water head H and vacuum value p_{v2} in the vacuum chamber.

Solution: Select cross section, datum plane 0-0, and let α=1

(1) Solve for *H*

Bernoulli equation for flow confluence:

$$E_1 \rho g q_{V1} + E_4 \rho g q_{V2} = E_3 \rho g q_V$$

where $E_1 = H \qquad E_4 = -h = -5 \text{ m}$

$$E_3 = \frac{1}{2g}\left(\frac{q_{V1}+q_{V2}}{D^2\pi/4}\right)^2$$

$$= \frac{1}{2\times 9.8}\left(\frac{0.0045}{(0.05)^2\times\pi/4}\right)^2 = 0.268 \text{ m}$$

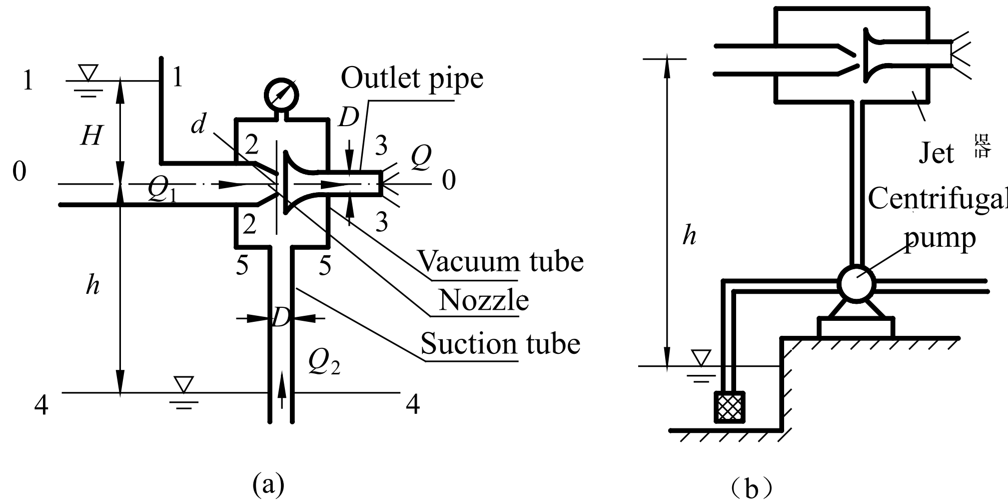


Fig. 4-14. Working principle and application of jet device

Then $H = E_1 = \dfrac{E_3 q_V - E_4 q_{V2}}{q_{V1}} = \dfrac{(0.268)\times(0.0045)-(-5)\times(0.0005)}{0.004} = 0.927 \text{ m}$

Note: If doing the calculation based on the Bernoulli equation for single-pipe total flow, the Bernoulli equation on Sections 1 & 3 gives: $E_1 = E_3$

It can be further got that H=0.268 m, which is incorrect.

Q: Given flow rates q_{V1}=0.004 m^3/s, q_{V2}=0.0005 m^3/s, h=5m, D=0.05m. Ignoring head loss, calculate the water head H and vacuum value p_{v2} in the vacuum chamber.

(2) Solve for p_{v2}

The Bernoulli equation (for unit-weight fluid) at sections 4-4 and 5-5 reads:

$$-h = \frac{1}{2g}\left(\frac{q_{V2}}{A_5}\right)^2 + \frac{p_2}{\rho g}$$

Re-organization gives:

$$\frac{p_2}{\rho g} = -\left[h + \frac{1}{2g}\left(\frac{q_{V2}}{D^2\,\pi/4}\right)^2\right]$$

$$= -\left[5 + \frac{1}{2\times 9.8}\left(\frac{0.0005}{(0.05)^2\times\pi/4}\right)^2\right] = -5.0033 \text{ m}$$

$p_{v2} = -p_2 =$ 5.0033 mH_2O

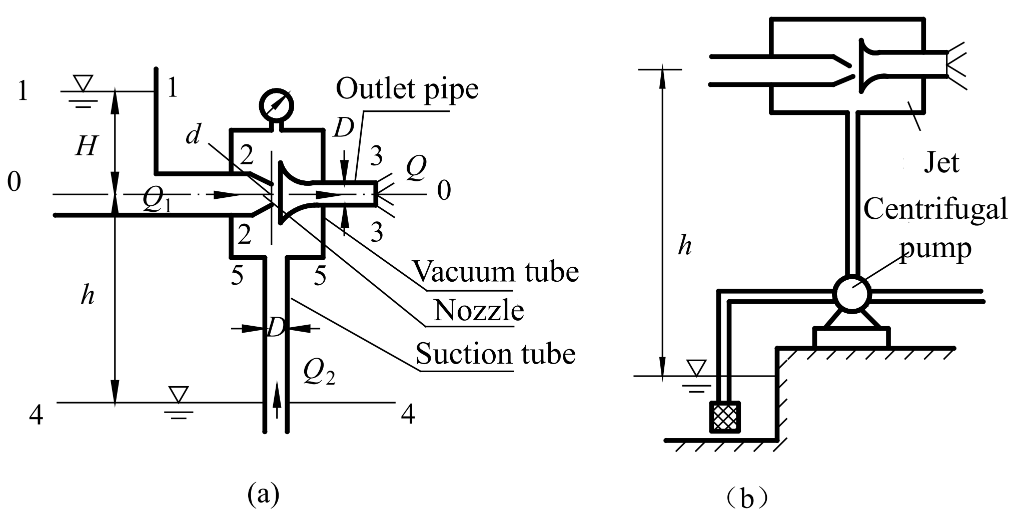


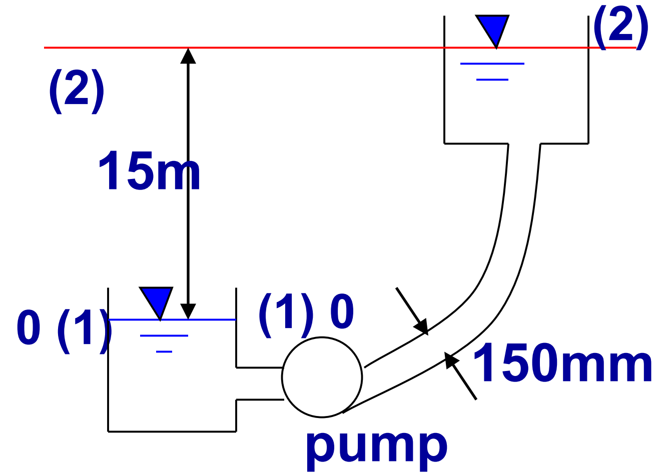
Fig. 4-14. Working principle and application of jet device

Example 2: A pumping system is transporting water from a lower pool to a higher one. The height difference between the two pools is 15 m; flow rate q_V=0.03m^3/s; the inner diameter of the pipe is d=150 mm. Efficiency of pump η_p=0.76. Given the head loss of the pipeline is $10v^2/(2g)$, solve for the shaft power of the pump.

Solution: Select cross section, datum plane

Letting the energy input of water (per unit weight) from the pump be h_p and $\alpha_1 = \alpha_2 = 1$, the Bernoulli equation at sections 1 & 2 reads:

$$z_1 + \frac{p_1}{\rho g} + \frac{v_1^2}{2g} + h_p = z_2 + \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + h_{w1,2}$$



Conditions: z_1=0, z_2=15m; p_1= p_2=0; $v_1 \approx v_2 \approx 0$ (because the flow section is large); the flow rate in the tube is:

$$v = \frac{q_V}{A} = \frac{0.03}{\frac{\pi}{4}(0.15)^2} = 1.7\text{m/s}$$

So: $$0+0+0+h_p = 15+0+0+10\frac{1.7^2}{19.6} \Longrightarrow$$ **h_p =16.47 m**

The required shaft power N_p is $$N_p = \frac{\rho g q_V h_p}{\eta_p} = \frac{9.8\times0.03\times16.47}{0.76} = 6.37\text{ kW}$$

Example 3: A natural smoke boiler, chimney diameter d = 1 m, smoke flow rate q_V=10m^3/s, smoke density ρ =0.7kg/m^3, external air density ρ_a =1.2kg/m^3, pressure loss of chimney $p_w = 0.035\frac{H}{d}\frac{\rho v^2}{2}$. The area of the inlet section at the bottom of the chimney is twice of the chimney, and the vacuum degree is 10 mm water column. Solve for the height of the chimney H.

Solution: Select cross section, datum plane. Since the densities of flue gas and external air are different, the Bernoulli equation for airflow is used, as:

$$p_1 + \frac{\rho v_1^2}{2} + (\rho_a - \rho) g (z_2 - z_1) = p_2 + \frac{\rho v_2^2}{2} + p_w$$

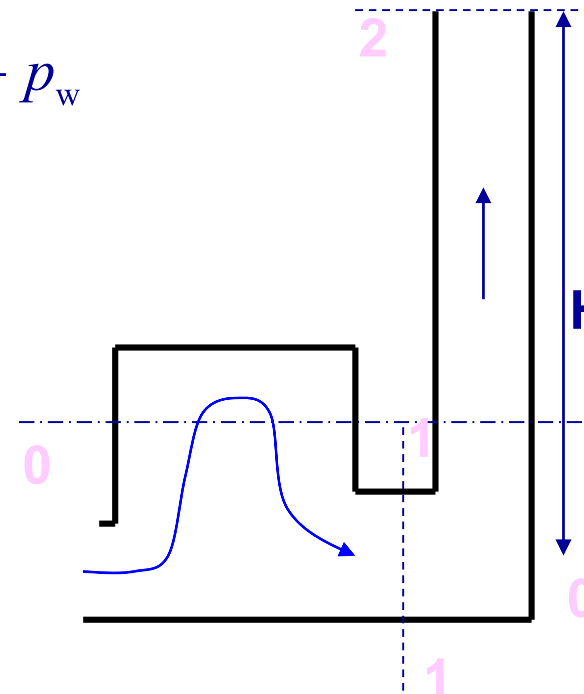
Section 1-1:

$$p_1 = -\rho_0 g h = -1000 \times 9.8 \times 0.01 = -98 \text{ N/m}^2$$

Also $z_1 = 0$; $v_1 A_1 = v_2 A_2 \longrightarrow v_1 = \frac{1}{2} v_2$

Section 2-2:

$$p_2 = 0,\ v_2 = \frac{q_V}{A} = 12.74 \text{ m/s},\ z_2 = H$$



Natural smoke boiler

Substitution gives:

H = 48.29 m, and $p_w = 0.035\frac{H}{d}\frac{\rho v_2^{\ 2}}{2} = 0.156H$

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Chapter 4 Basic equations of steady total flow

Section 1 Method of total flow analysis

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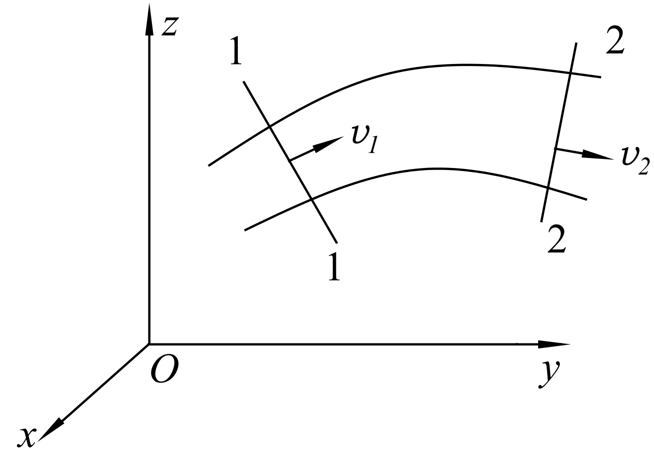
☞ Section 4 Momentum equation for steady total flow

Chapter summary

Section 4 Momentum equation for steady total flow

The law of momentum specifies that the temporal variation rate of momentum of a system is equal to the vector sum of all external forces acting on the system. That is,



$$\sum \vec{F} = \frac{\mathrm{d}\vec{K}}{\mathrm{d}t} = \frac{\mathrm{d}(\sum m\vec{u})}{\mathrm{d}t}$$

$$\vec{K}_1 = \beta_1 \rho q_V \vec{\upsilon}_1 \cdot \Delta t; \quad \vec{K}_2 = \beta_2 \rho q_V \vec{\upsilon}_2 \cdot \Delta t$$

Review

The momentum of element flow through the cross section within time Δt:

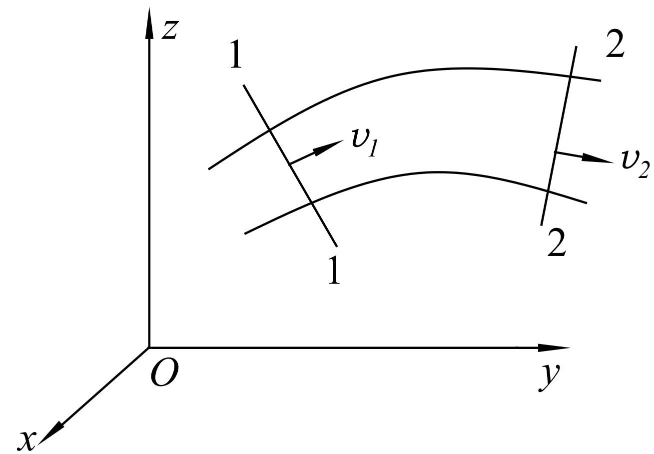
$$\mathrm{d}\vec{K} = \mathrm{d}m\vec{u} = (\rho u \mathrm{d}A \cdot \Delta t)\vec{u} = \rho u \vec{u} \mathrm{d}A \cdot \Delta t$$

The momentum of the total flow through the whole cross section is:

$$\sum \mathrm{d}\vec{K} = \int_A \rho u \vec{u} \mathrm{d}A \cdot \Delta t = [\int_A \rho u \mathrm{d}A u]\vec{i} \cdot \Delta t = \beta \rho q_V \upsilon \cdot \vec{i} \cdot \Delta t = \beta \rho q_V \vec{\upsilon} \cdot \Delta t$$

Section 4 Momentum equation for steady total flow

The law of momentum specifies that the temporal variation rate of momentum of a system is equal to the vector sum of all external forces acting on the system. That is,



$$\sum \vec{F} = \frac{\mathrm{d}\vec{K}}{\mathrm{d}t} = \frac{\mathrm{d}(\sum m\vec{u})}{\mathrm{d}t}$$

$$\vec{K}_1 = \beta_1 \rho q_V \vec{\upsilon}_1 \cdot \Delta t; \quad \vec{K}_2 = \beta_2 \rho q_V \vec{\upsilon}_2 \cdot \Delta t$$

Then $\sum \vec{F} = \dfrac{\vec{K}_2 - \vec{K}_1}{\Delta t} = \rho q_V (\beta_2 \vec{\upsilon}_2 - \beta_1 \vec{\upsilon}_1)$

Writing in the form of components in a Cartesian coordinate system gives:

$$\begin{cases} \sum F_x = \rho q_V (\beta_2 \upsilon_{2x} - \beta_1 \upsilon_{1x}) \\ \sum F_y = \rho q_V (\beta_2 \upsilon_{2y} - \beta_1 \upsilon_{1y}) \\ \sum F_z = \rho q_V (\beta_2 \upsilon_{2z} - \beta_1 \upsilon_{1z}) \end{cases}$$

This is the momentum equation for steady total flow of incompressible fluid.

External forces acting on the fluid within a control body $=$ Net outflow momentum (difference between outflow and inflow momentum) of the control volume per unit time, along the direction of external force

$$\sum \vec{F} = \rho q_V \ (\beta_2 \vec{v}_2 - \beta_1 \vec{v}_1)$$

录像1

where, $\sum \vec{F}$: The vector sum of all external forces acting on the fluid within the control volume, including :

1) The mass force acting on all fluid particles with in the control volume;

2) All surface forces acting on the surfaces of the control volume (e.g. shear force, hydrodynamic pressure)

3) The total force acted by the surrounding boundaries

Scope of application:

1) Ideal fluid, real fluid, incompressible fluid, steady flow

2) Two flow cross sections selected must be gradually-varied or uniform flow sections, while flows between the two sections can be rapidly-varied ones

3) Gravity being the sole mass force

4) Rate of total flow remains unchanged along the direction of flow motion;

5) If the flow rate changes, the equation becomes:

$$\sum (\rho q_V \beta \vec{v})_{\text{in}} - \sum (\rho q_V \beta \vec{v})_{\text{out}} = \sum \vec{F}$$

Solution steps by using momentum equation:

1. Select control volume: According to requirements, take the water body between two gradually-varied-flow sections as the control volume;
2. Define coordinate system, and determine the amplitudes and directions of forces and projected velocities;
3. Plot the free-body diagram by indicating the directions of all forces, and analyze the forces on the water body;
4. Write momentum equations, and solve for unknowns by substituting given forces and velocities.

Notes: 1) the relative pressure is used in the calculation;

2) The Bernoulli equation and continuity equation can be incorporated in the calculation

Example 1: Overflow dam. The upstream water level is raised due to the blocking of the dam. The water depth at section 1-1 is 1.5 m, and that at the downstream section 2-2 is 0.6 m. Ignoring the head loss, calculate the horizontal force F_R of water flow on a 1 m segment (perpendicular to the paper sheet) of the dam.

Solution: 1) Take the control volume and define the coordinate system

2) Force analysis

$$F_x = F_{P1} - F_{P2} - F_R{}' = \frac{\rho g}{2} H_1^2 - \frac{\rho g}{2} H_2^2 - F_R{}'$$

3) The momentum equation

$$\frac{\rho g}{2} H_1^2 - \frac{\rho g}{2} H_2^2 - F_R{}' = \rho q_V (v_2 - v_1)$$

4) Supplementary conditions for v_1 and v_2

Continuity equation: $q_V = H_1 v_1 = H_2 v_2$

5) The total flow Bernoulli equation

$$H_1 + \frac{v_1^2}{2g} = H_2 + \frac{v_2^2}{2g}$$

6) Combining the above equations gives:

$F_R{}'$ = 1.71 kN (direction consistent with that in the diagram)

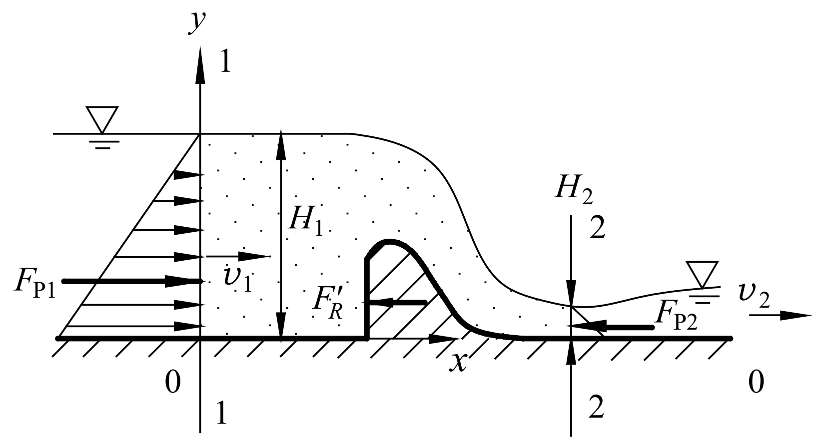


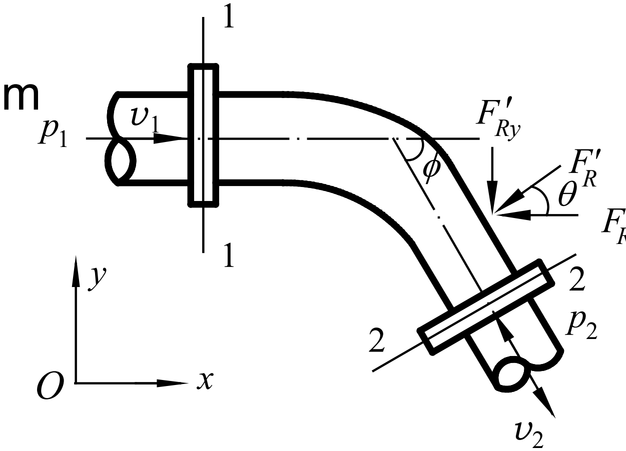
Fig 4-18 Force of water flow on dam

Note: The force on the dam applied by water flow equals to that on the water flow acted by the dam, and the directions are opposite. That is,

F_R=1.71 kN

Direction opposite to that in the figure

Example 2 : A horizontally deployed pipeline with 60° bend, the inner diameter d_1=1 m, d_2=0.75 m, flow rate q_V=1.57m^3/s, the pressure on the elbow inlet section p_1=5.05×10^4 Pa. Ignoring the head loss, solve for the horizontal thrust applied on the elbow by the water flow.



Solution: (1) Take control volume; select coordinate system

(2) Force analysis

$$F_x = p_1 \frac{\pi}{4} d_1^2 - p_2 \frac{\pi}{4} d_2^2 \cos\phi - F_{Rx}'$$

$$F_y = p_2 \frac{\pi}{4} d_2^2 \sin\phi - F_{Ry}'$$

(3) Write momentum equation for F_{Rx}' F_{Ry}'

x: $p_1 \frac{\pi}{4} d_1^2 - p_2 \frac{\pi}{4} d_2^2 \cos\phi - F_{Rx}' = \rho q_V (v_2 \cos\phi - v_1)$

y: $p_2 \frac{\pi}{4} d_2^2 \sin\phi - F_{Ry}' = \rho q_V (-v_2 \sin\phi - 0)$

(4) Supplementary conditions for $\boldsymbol{v_1}$、$\boldsymbol{v_2}$ and $\boldsymbol{p_2}$

$$v_1 = \frac{4q_V}{\pi d_1^2} = 2 \text{ m/s}; \quad v_2 = \frac{4q_V}{\pi d_2^2} = 3.56 \text{ m/s}$$

$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + 0$$

$\Longrightarrow$ p_2 = 4.616×10^4 Pa

Combining (2) to (4) gives:

F_{Rx}' = 29.8 kN F_{Ry}' = 22.5 kN

$F_R = \sqrt{F_{Rx}'^2 + F_{Ry}'^2} = 37.3 \text{ kN}$, opposite to F_R'

$$\theta = \arctan\frac{F_{Ry}}{F_{Rx}} = 37.1°$$

Example 3: A dam with the water depth in front being H=150 m. The energy dissipation approach of high and low jet-flow collision is adopted for flooding discharge. Assuming the flow rates of the two jet flows being equal, do the following hydraulic design calculations:

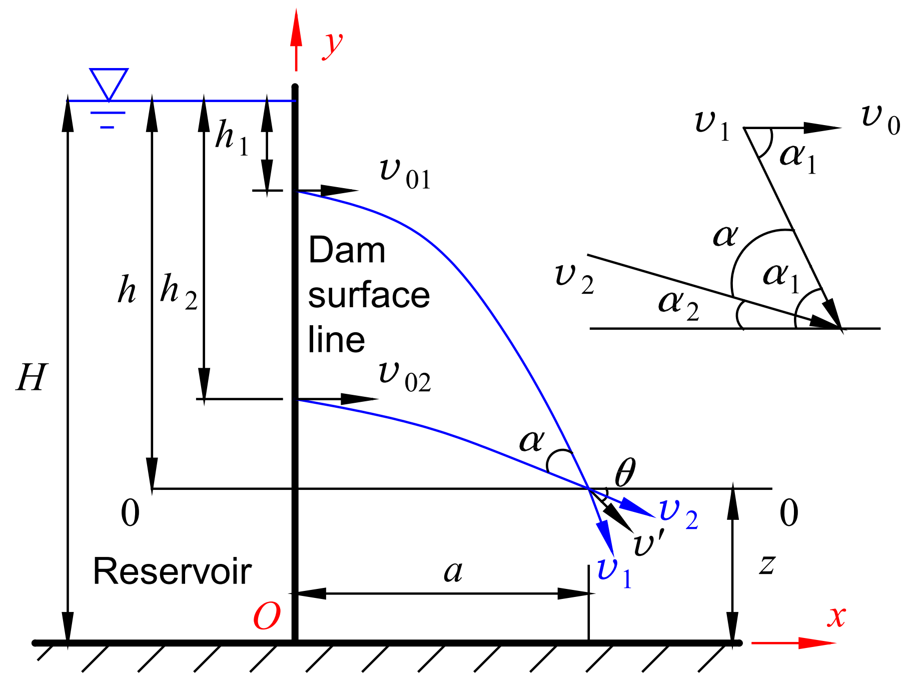
(1) If the collision location is a = 50 m from the dam surface line and z = 50 m from the dam toe, solve for the depths of the two holes, i.e. h_1 and h_2;

Solution:

Horizontal projectile motions gives:

$$\begin{cases} \upsilon_{01} t_1 = a \\ \upsilon_{02} t_2 = a \end{cases} \qquad \begin{cases} h - h_1 = \dfrac{1}{2} g t_1^2 \\ h - h_2 = \dfrac{1}{2} g t_2^2 \end{cases}$$

Ignoring head loss, we have: $\begin{cases} \upsilon_{01} = \sqrt{2gh_1} \\ \upsilon_{02} = \sqrt{2gh_2} \end{cases}$



$$\Longrightarrow \quad h_1 = \frac{h - \sqrt{h^2 - a^2}}{2} \qquad h_2 = \frac{h + \sqrt{h^2 - a^2}}{2}$$

With h = 100 m， a = 50 m, substituting gives:

h_1 = 6.7 m， h_2 = 93.3 m

Example 3: A dam with the water depth in front being H=150 m. The energy dissipation approach of high and low jet-flow collision is adopted for flooding discharge. Assuming the flow rates of the two jet flows being equal, do the following hydraulic design calculations:

(2) Find the collision angle of two jet flows, i.e. α;

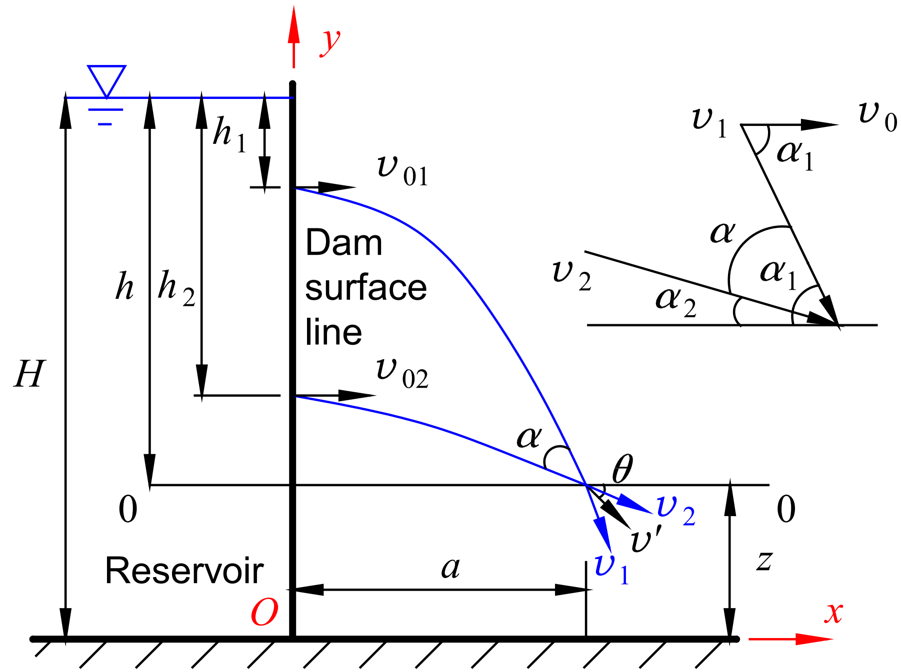
Solution:

From the figure, $\alpha = \alpha_1 - \alpha_2$

Ignoring head loss, $v = v_1 = v_2 = \sqrt{2gh}$

$$\cos\alpha_1 = \frac{v_{01}}{v_1} = \frac{\sqrt{2gh_1}}{\sqrt{2gh}} = \frac{\sqrt{h_1}}{\sqrt{h}}$$

$$\cos\alpha_2 = \frac{\sqrt{h_2}}{\sqrt{h}}$$

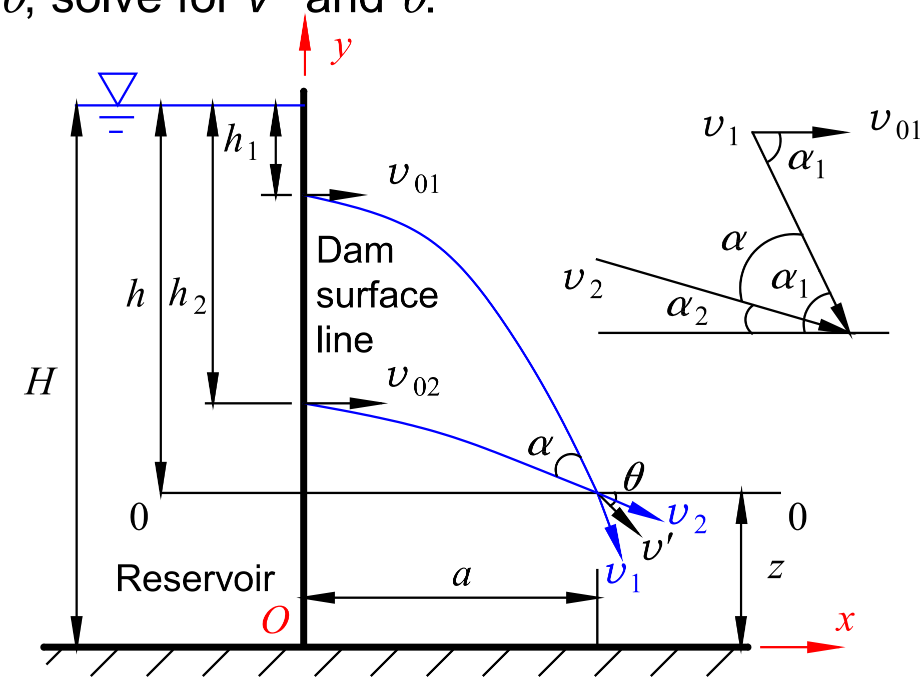


Substituting h=100 m, h_1=6.7 m, h_2=93.3 m gives:

$$\alpha_1 = 75°, \alpha_2 = 15°, \alpha = 60°$$

Example 3: A dam with the water depth in front being H=150 m. The energy dissipation approach of high and low jet-flow collision is adopted for flooding discharge. Assuming the flow rates of the two jet flows being equal, do the following hydraulic design calculations:

(3) After the collision, the main stream flows in the velocity v' and in the direction of intersecting with the horizontal plane by θ, solve for v' and θ.



Solution: In the coordinate system Oxy, write the momentum equation:

$$F_x = 2\beta\rho q_V \upsilon' \cos\theta -$$
$$(\beta_1 \rho q_V \upsilon \cos\alpha_1 + \beta_1 \rho q_V \upsilon \cos\alpha_2) = 0$$

$$F_y = 2\beta\rho q_V \upsilon' \sin\theta -$$
$$\beta_1 \rho q_V \upsilon (\sin\alpha_1 + \sin\alpha_2) = 0$$

$$\Longrightarrow \tan\theta = \frac{\sin\alpha_1 + \sin\alpha_2}{\cos\alpha_1 + \cos\alpha_2} = 1 \quad \Rightarrow \theta = 45^\circ$$

$$\upsilon' = \frac{\upsilon(\cos\alpha_1 + \cos\alpha_2)}{2\cos\theta} = \frac{\sqrt{2\times 9.8\times 100}\times(\cos 15^\circ + \cos 75^\circ)}{2\cos 45^\circ}$$

$$= 38.36 \text{ m/s} \quad (\text{and } \upsilon = 44.3 \text{ m/s})$$

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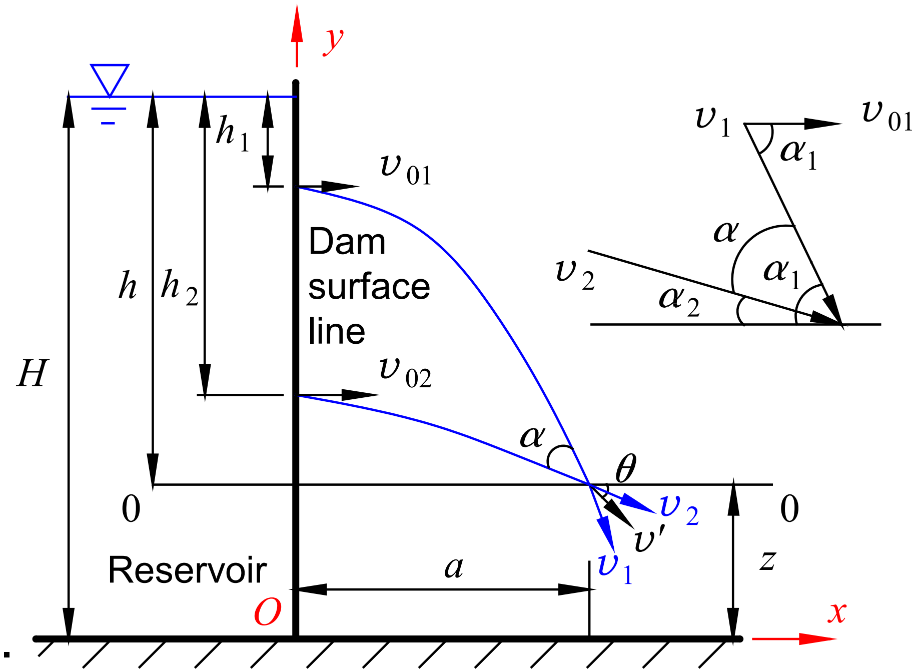
Example 3: A dam with the water depth in front being H=150 m. The energy dissipation approach of high and low jet-flow collision is adopted for flooding discharge. Assuming the flow rates of the two jet flows being equal, do the following hydraulic design calculations:

(4) Calculate the energy dissipation efficiency η induced by jet collision. (Ignore head losses other than collision, and ignore the effect of gravity during collision)

Solution:

$$\eta = \frac{\frac{v^2}{2g} - \frac{v'^2}{2g}}{\frac{v^2}{2g}} = 1 - \left(\frac{38.36}{44.3}\right)^2 = 0.25$$



The energy loss caused by the collision of two jet flows equals to 25% of the total energy of the flows before collision. That is, the energy dissipation efficiency is 25 %.

1. Why does the momentum equation for steady total flow not contain the head loss term explicitly?

A: The head loss has actually been considered. This is because the external force *F* in the momentum equation includes the shear force, and the shear stress is directly related to the head loss.

2. (1) What does the negative force obtained from the momentum equation imply? (2) Are the magnitudes of unknowns force related to the size of isolated body (control volume)? (3) How to select the isolated body in practice?

A: (1) Opposite to the defined positive direction; (2) Unrelated in the case without gravity; (3) The calculation sections and solid boundaries can be selected.

3. 'The values of $p/\rho g$ at different locations on a gradually-varied flow section are constant. Is this statement correct or not? Why?

A: No. The piezometric head is constant, i.e. $z+\frac{p}{\rho g}=C$

Since Z is different, the pressure head (or called piezometric height) is different, i.e. $\frac{p}{\rho g}\neq C$

Chapter 4 Basic equations of steady total flow

Section 1 Method of total flow analysis

Section 2 Continuity equation

Section 3 Bernoulli equation for steady total flow

Section 4 Momentum equation for steady total flow

☞ Chapter summary

Chapter summary

1. Several basic concepts

1）Element flow & total flow

2）Cross section of flow: the cross section perpendicular to the flow direction in an channel (e.g. pipeline, open channel, etc.), or the cross section orthogonal to the streamline of element or total flows.

3）Point velocity: The velocity at a point of fluid flow (denoted by ***u***). In general, the point velocity at different locations on a flow section is not equal.

4）Mean velocity: $v = \dfrac{\int_A u \mathrm{d}A}{A}$

5）Gradually-varied flow: A flow whose streamlines are almost parallel straight lines. Or a flow with curved streamlines but radius of curvature is very large. The limit of gradually-varied flow is uniform flow. The distribution of dynamic pressures on a gradually-varied-flow section is the same as that of hydrostatic pressure, i.e., $z+p/\rho g$ is constant.

However it is noted that $z+p/\rho g$ is generally not equal on different sections.

6）Rapidly-varied flow: Opposite to the characteristics of gradually-varied flow.

7）Correction factors for momentum and kinetic energy

2. Continuity equation of steady total flow

Incompressible fluid flow without bifurcation:

$$v_1 A_1 = v_2 A_2 \text{ or } q_{V1} = q_{V2}$$

which implies that the average velocity of a certain cross section is inversely proportional to the area of the section.

Incompressible fluid flow with bifurcation:

$$\sum q_{V\text{-in}} = \sum q_{V\text{-out}}$$

which implies that the sum of inflow to the bifurcation point equals to that of the outflow from the bifurcation point.

3. Bernoulli equation for steady total flow

(1) Bernoulli equation

$$z_1+\frac{p_1}{\rho g}+\frac{\alpha_1 v_1^2}{2g}=z_2+\frac{p_2}{\rho g}+\frac{\alpha_2 v_2^2}{2g}+h_{\mathrm{w}}$$

	Physical meaning	Geometric meaning
z —	Specific potential energy at the calculation point on the total-flow cross section	Elevation head
$\frac{p}{\rho g}$ —	Specific pressure energy	Pressure head
$\frac{u^2}{2g}$ —	Specific kinetic energy	Flow head
$z+\frac{p}{\rho g}$ —	Total specific potential energy	Hydraulic (piezometric) head
$z+\frac{p}{\rho g}+\frac{u^2}{2g}$ —	Total specific energy	Total head

(2) Application conditions of Bernoulli equation

(3) Notes when applying Bernoulli equation

1) Take flow sections in gradually-varied flow areas along flow direction;

2) The datum can be arbitrary in theory, but better to select such that the water heads at sections be positive;

3) The same pressure reference level must be adopted in the same problem. Relative pressure is generally used. Better to use absolute pressure when vacuum happens.

4) Since $Z+p/\rho g$ is constant, the calculation point can be arbitrarily taken on the section. For pipe flows, the center point of the section is often taken; for the open channel flow, the calculation point is often taken on the free surface;

5) Better to select sections with more known quantities, e.g. sections in the upstream pool where v_1=0 and p=0, and the downstream pipeline outlet section with p_2 = 0. The selected sections should include the unknown quantities to solve;

6) When there are 2 - 3 unknowns in a problem, the continuity and momentum equations are needed;

7) The Bernoulli equation can be applied to flows with bifurcation, because the above Bernoulli equation is for fluid of unit weight;

8) When there are energy input and/or output at sections, the Bernoulli equation reads:

$$z_1+\frac{p_1}{\rho g}+\frac{\alpha_1 v_1^2}{2g}\pm H=z_2+\frac{p_2}{\rho g}+\frac{\alpha_2 v_2^2}{2g}+h_w$$

(4) Head lines

4. Momentum equation for steady total flow

$$\begin{cases} \sum F_{\mathrm{x}} = \rho q_V(\beta_2 v_{2\mathrm{x}} - \beta_1 v_{1\mathrm{x}}) \\ \sum F_{\mathrm{y}} = \rho q_V(\beta_2 v_{2\mathrm{y}} - \beta_1 v_{1\mathrm{y}}) \\ \sum F_{\mathrm{z}} = \rho q_V(\beta_2 v_{2\mathrm{z}} - \beta_1 v_{1\mathrm{z}}) \end{cases}$$

The projection of the resultant external force (acted on a flow segment) on a certain coordinate axis equals to the difference in momentum flowing out of and into the flow segment in that direction.

For bifurcated pipe flow, the momentum equation reads:

$$\sum \vec{F} = (\rho q_V \beta \vec{v})_{\mathbf{out}} - (\rho q_V \beta \vec{v})_{\mathbf{in}}$$

Some notes:

$$\begin{cases} \sum F_{\mathrm{x}} = \rho q_V(\beta_2 v_{2\mathrm{x}} - \beta_1 v_{1\mathrm{x}}) \\ \sum F_{\mathrm{y}} = \rho q_V(\beta_2 v_{2\mathrm{y}} - \beta_1 v_{1\mathrm{y}}) \\ \sum F_{\mathrm{z}} = \rho q_V(\beta_2 v_{2\mathrm{z}} - \beta_1 v_{1\mathrm{z}}) \end{cases}$$

1） The isolated free body should be taken at gradually-varied flow sections, but the flow within it can be rapidly-varied flows;

2） Momentum equations are in the vector form. Proper projection axes should be selected. Pay attention to the signs of force and velocity;

3） The external forces include the mass force and surface force acting on the free body. The direction of the force acting on the fluid by the solid boundary can be assumed beforehand. If the calculated force value is positive, the assumed direction is correct; otherwise, the actual force direction is opposite to the assumed one;

4） It should be momentum output minus momentum input;

5） The momentum equation can only solve for one unknown. If there is more than one unknown, the continuity equation and Bernoulli equation should be combined.